Evaluation of CSP methods

Soundness and completeness of algorithms

Definition 2-1:

An algorithm is **sound** if every result that is returned by it is indeed a solution; in CSPs, that means any compound label which is returned by it contains labels for every variable, and this compound label satisfies all the constraints in the problem. ■

Definition 2-2:

An algorithm is **complete** if every solution can be found by it. ■

Accuracy vs Precision vs Recall

Accuracy refers to the closeness of a measured value to a standard or known value.

For example, if in lab you obtain a weight measurement of 3.2 kg for a given substance, but the actual or known weight is 10 kg, then your measurement is not accurate. In this case, your measurement is not close to the known value.

Precision refers to the closeness of two or more measurements to each other.

Using the example above, if you weigh a given substance five times, and get 3.2 kg each time, then your measurement is very precise. Precision is independent of accuracy. You can be very precise but inaccurate, as described above. You can also be accurate but imprecise.

For example, if on average, your measurements for a given substance are close to the known value, but the measurements are far from each other, then you have accuracy without precision.

Another example: imagine a basketball player shooting baskets.

If the player shoots with accuracy, his aim will always take the ball close to or into the basket.

If the player shoots with precision, his aim will always take the ball to the same location which may or may not be close to the basket.

A good player will be both accurate and precise by shooting the ball the same way each time and each time making it in the basket.

$$\text{Precision} = \frac{tp}{tp + fp} \qquad \text{True negative rate} = \frac{tn}{tn + fp}$$

$$\text{Recall} = \frac{tp}{tp + fn} \qquad \text{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn}$$

condition positive (P)	the number of real positive cases in the data
condition negatives (N)	the number of real negative cases in the data
true positive (TP)	case positive and predicted positive
true negative (TN)	case negative and predicted negative
false positive (FP)	case negative and predicted positive
false negative (FN)	case positive and predicted negative

What percent of predictions were correct?

The "accuracy"

What percent of positive cases did you catch?

The "recall"

What percent of positive predictions were correct?

The "precision"